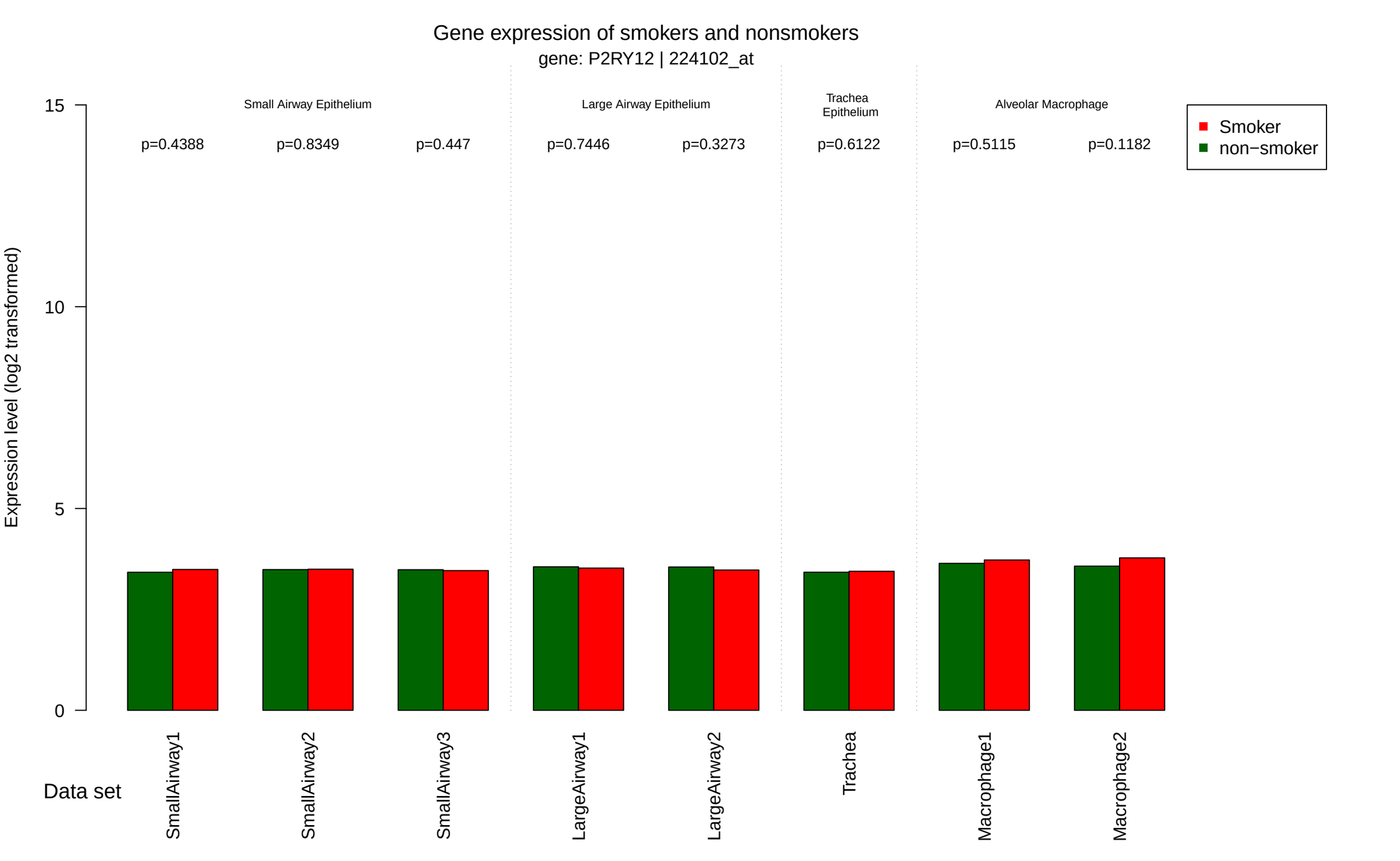
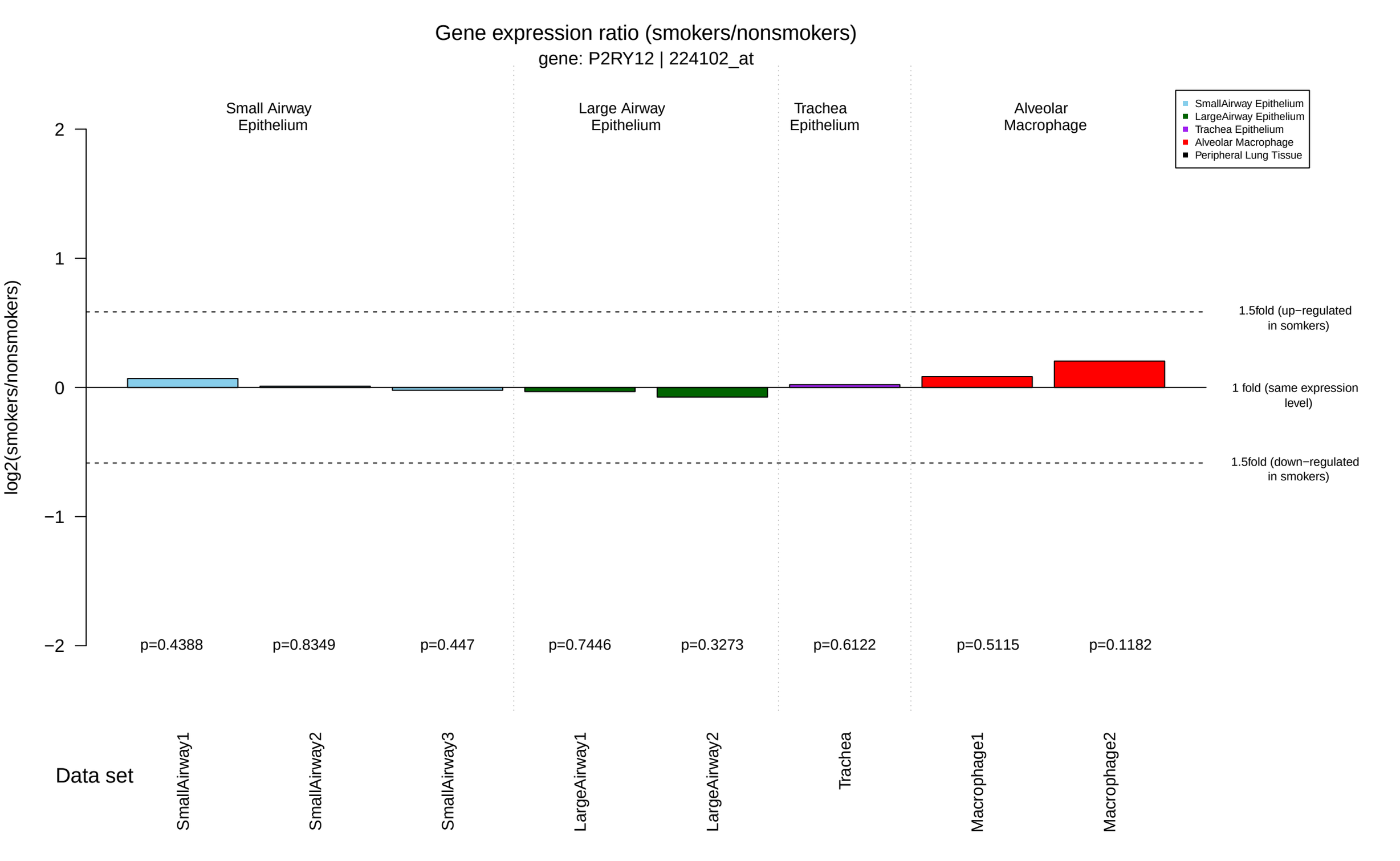
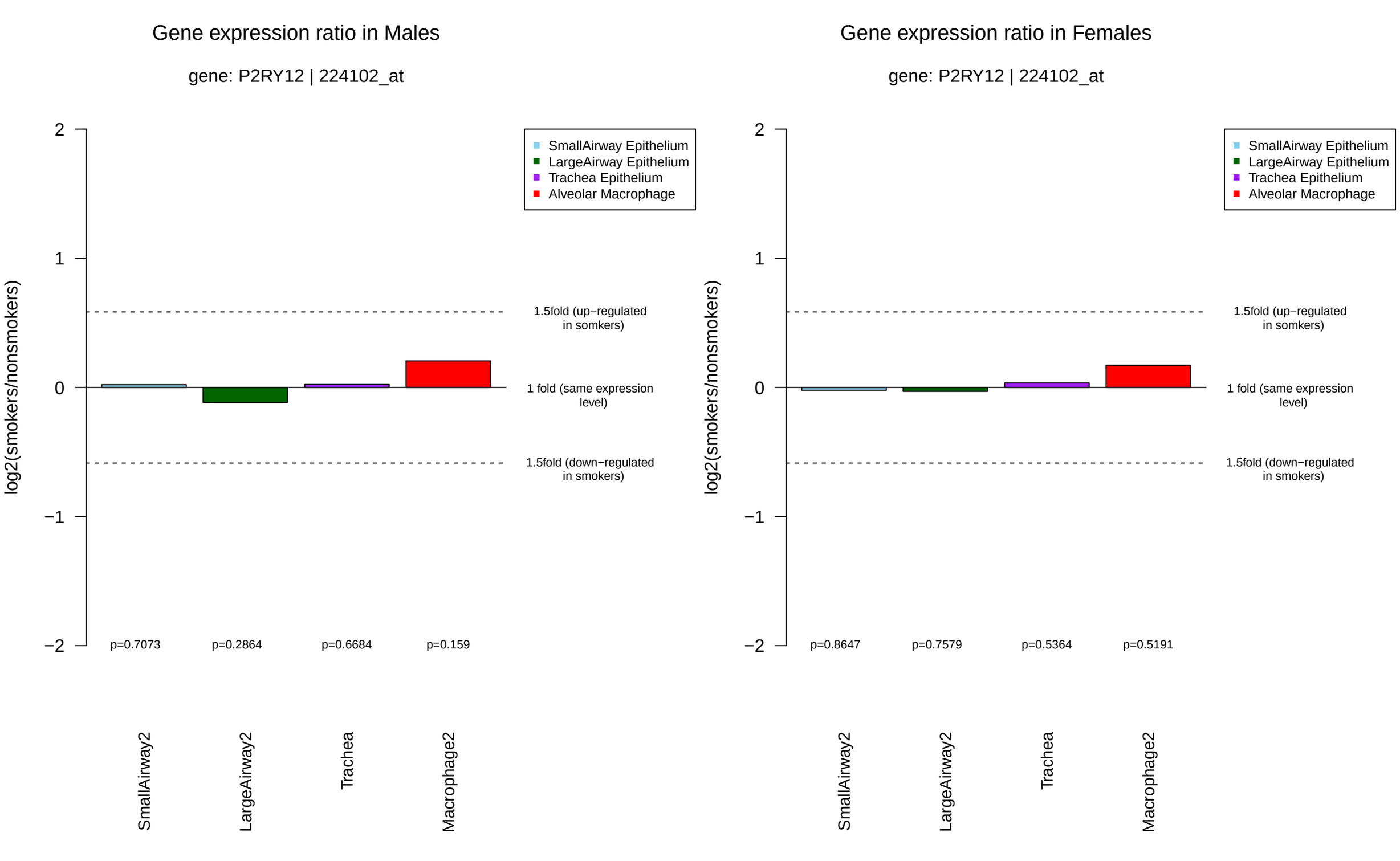
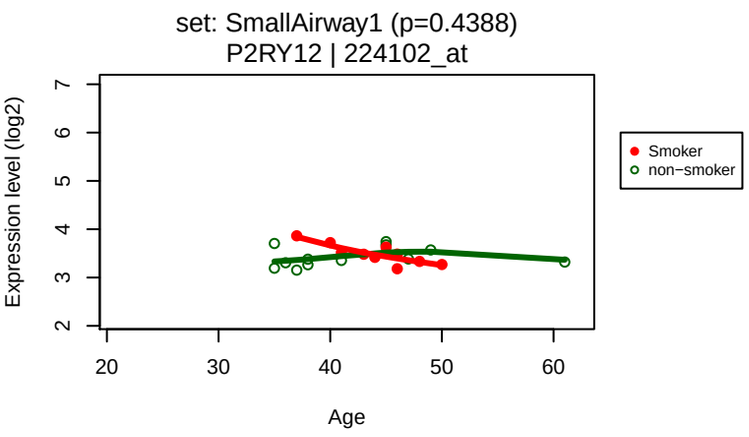
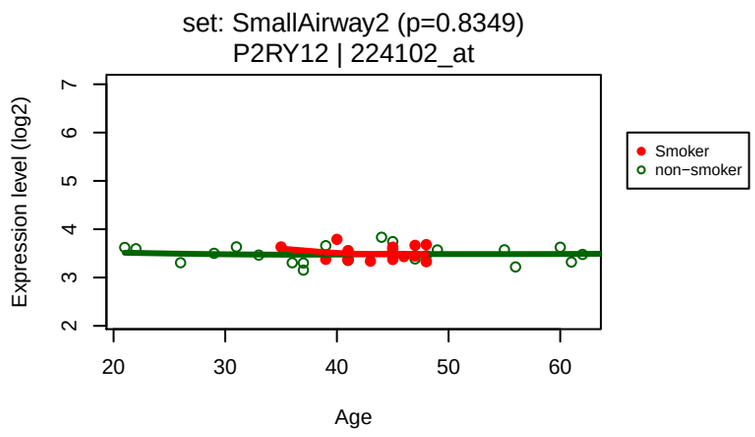
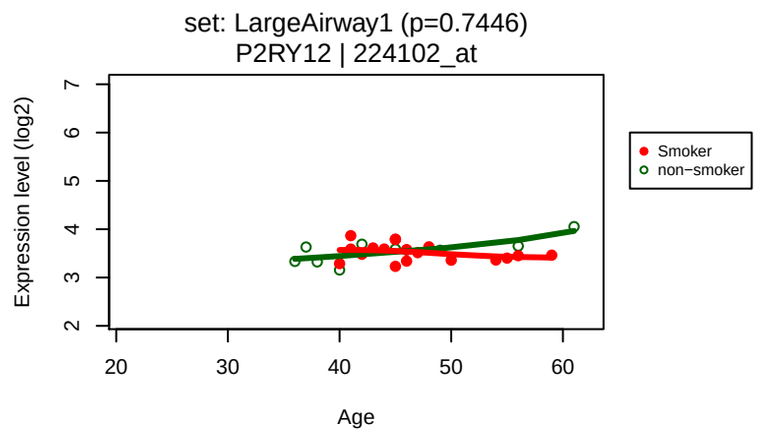

Gene expression of smokers and nonsmokers
gene: P2RY12 | 224102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4388
p=0.8349
p=0.447
p=0.7446
p=0.3273
p=0.6122
p=0.5115
p=0.1182
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

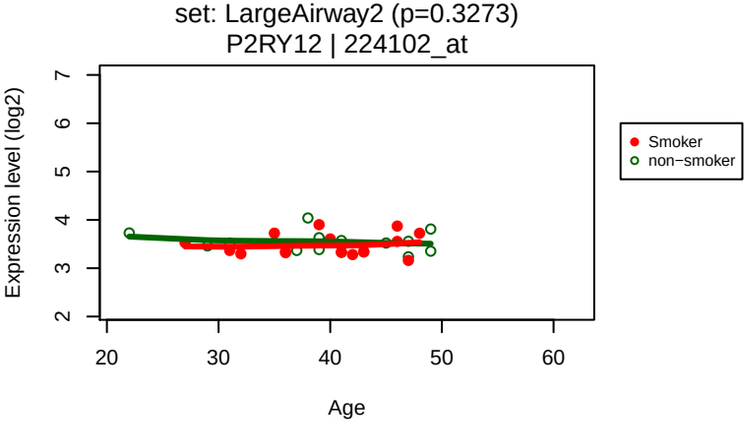

Gene expression ratio (smokers/nonsmokers)
gene: P2RY12 | 224102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4388
p=0.8349
p=0.447
p=0.7446
p=0.3273
p=0.6122
p=0.5115
p=0.1182
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

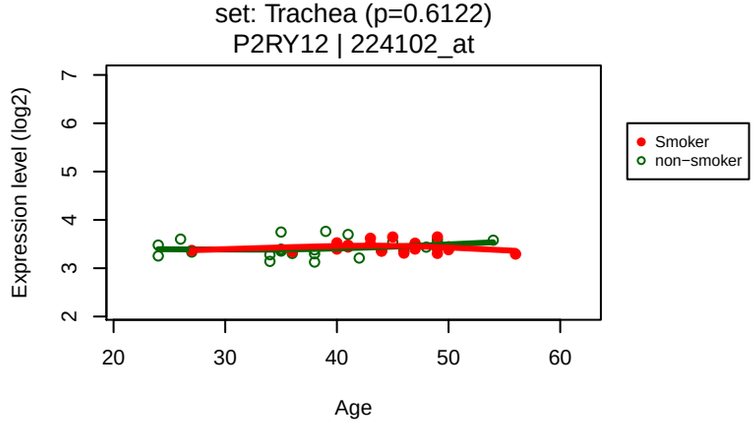

Gene expression ratio in Males
gene: P2RY12 | 224102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7073
p=0.2864
p=0.6684
p=0.159
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: P2RY12 | 224102_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8647
p=0.7579
p=0.5364
p=0.5191
SmallAirway2
LargeAirway2
Trachea
Macrophage2

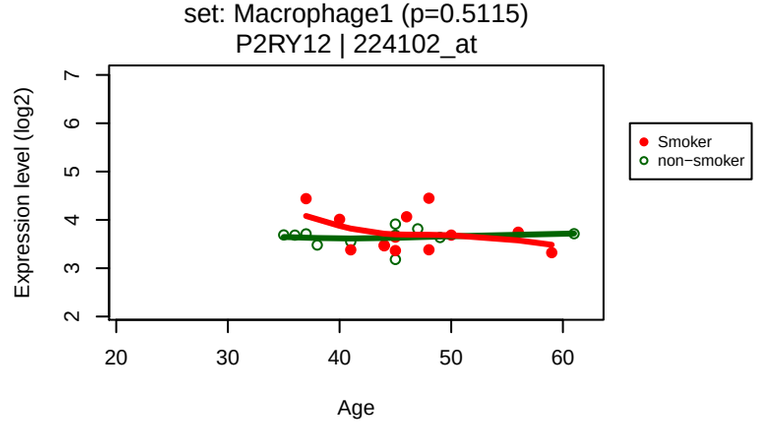

set: SmallAirway1 (p=0.4388)
P2RY12 | 224102_at
Expression level (log2)
Age
Smoker
non−smoker

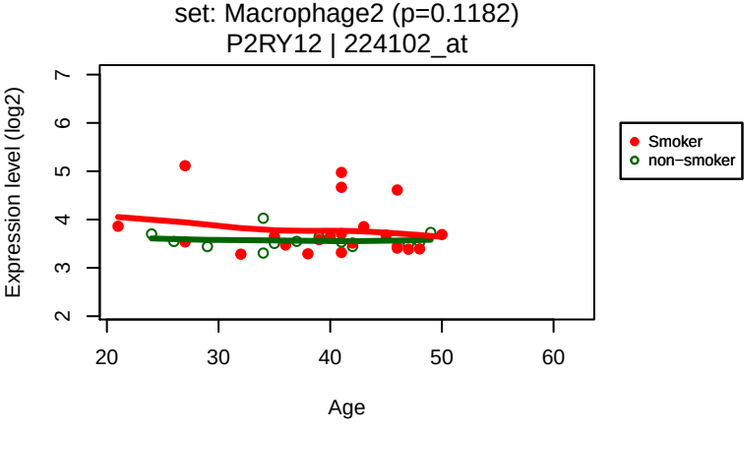

set: SmallAirway2 (p=0.8349)
P2RY12 | 224102_at
Expression level (log2)
Age
Smoker
non−smoker

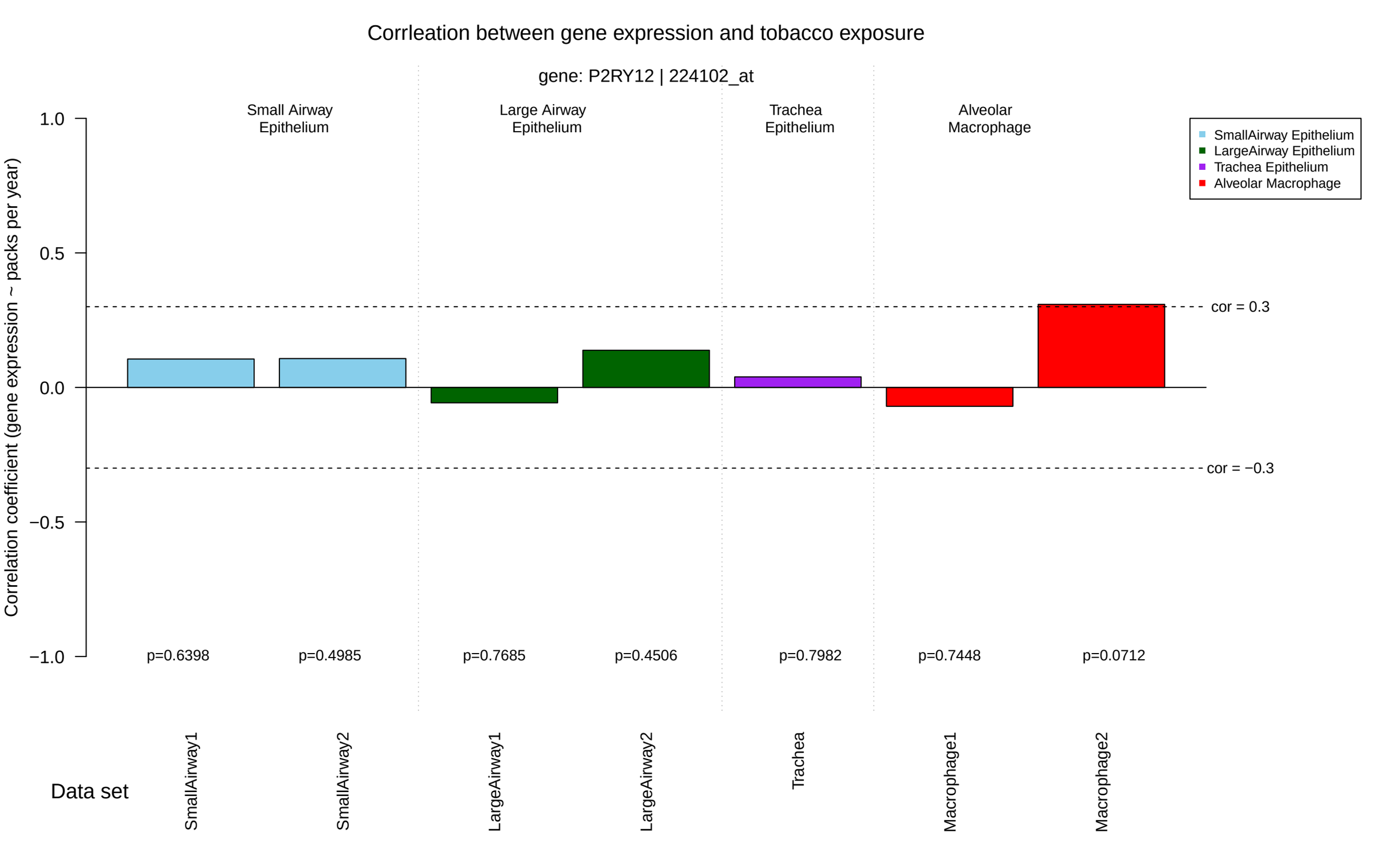

set: LargeAirway1 (p=0.7446)
P2RY12 | 224102_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3273)
P2RY12 | 224102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6122)
P2RY12 | 224102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5115)
P2RY12 | 224102_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1182)
P2RY12 | 224102_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: P2RY12 | 224102_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.6398
p=0.4985
p=0.7685
p=0.4506
p=0.7982
p=0.7448
p=0.0712
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2